$$
\begin{aligned}
&C_{\phi q}^{(12)[1]}\text{i1\_i2\_} \to \\
&\frac{1}{16\pi^2}\Bigg(\frac{1}{18}\tilde{\mu}\, g_1^2\, \overline{y_d}^{pr} a_d^{pr}\, \mathrm{LF}_{2,2,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}]\,\delta_{i1i2} - \frac{1}{36}\tilde{\mu}\, g_1^2\, \overline{y_d}^{pr} a_d^{pr}\, \mathrm{LF}_{3,2,-1}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}]\,\delta_{i1i2} + \\
&\frac{1}{18}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pr} a_e^{pr}\, \mathrm{LF}_{2,2,0}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}]\,\delta_{i1i2} - \frac{1}{36}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pr} a_e^{pr}\, \mathrm{LF}_{3,2,-1}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}]\,\delta_{i1i2} - \\
&\frac{1}{18}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pr} a_e^{pr}\, \mathrm{LF}_{3,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}]\,\delta_{i1i2} - \frac{1}{18}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pr} a_e^{pr}\, \mathrm{LF}_{3,2,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}]\,\delta_{i1i2} + \\
&\frac{1}{8}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pr} a_e^{pr}\, \mathrm{LF}_{4,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}]\,\delta_{i1i2} - \frac{1}{18}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pr} a_e^{pr}\, \mathrm{LF}_{5,1,-2}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}]\,\delta_{i1i2} - \\
&\frac{1}{18}\tilde{\mu}\, g_1^2\, \overline{y_d}^{pr} a_d^{pr}\, \mathrm{LF}_{3,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}]\,\delta_{i1i2} - \frac{1}{18}\tilde{\mu}\, g_1^2\, \overline{y_d}^{pr} a_d^{pr}\, \mathrm{LF}_{3,2,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}]\,\delta_{i1i2} + \\
&\frac{5}{24}\tilde{\mu}\, g_1^2\, \overline{y_d}^{pr} a_d^{pr}\, \mathrm{LF}_{4,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}]\,\delta_{i1i2} - \frac{1}{6}\tilde{\mu}\, g_1^2\, \overline{y_d}^{pr} a_d^{pr}\, \mathrm{LF}_{5,1,-2}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}]\,\delta_{i1i2} - \\
&\frac{1}{36}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pr} a_u^{pr}\, \mathrm{LF}_{2,2,0}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}]\,\delta_{i1i2} + \frac{1}{72}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pr} a_u^{pr}\, \mathrm{LF}_{3,2,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}]\,\delta_{i1i2} + \\
&\frac{1}{36}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pr} a_u^{pr}\, \mathrm{LF}_{3,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}]\,\delta_{i1i2} + \frac{1}{36}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pr} a_u^{pr}\, \mathrm{LF}_{3,2,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}]\,\delta_{i1i2} + \\
&\frac{1}{12}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pr} a_u^{pr}\, \mathrm{LF}_{4,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}]\,\delta_{i1i2} - \frac{1}{6}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pr} a_u^{pr}\, \mathrm{LF}_{5,1,-2}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}]\,\delta_{i1i2} + \\
&\frac{1}{24} m_1\tilde{\mu}\, g_1^4\, \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_1]\,\delta_{i1i2} - \frac{1}{18} m_1\tilde{\mu}\, g_1^4\, \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_1]\,\delta_{i1i2} + \\
&\frac{1}{8} m_2\tilde{\mu}\, g_1^2 g_2^2\, \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_2]\,\delta_{i1i2} - \frac{1}{6} m_2\tilde{\mu}\, g_1^2 g_2^2\, \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_2]\,\delta_{i1i2} + \frac{1}{72} m_1\tilde{\mu}\, g_1^2 \\
&\quad \overline{y_d}^{i2p} y_d^{i1p}\, \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}] + \frac{1}{144} m_1\tilde{\mu}\, g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}] - \\
&\frac{1}{72} m_1\tilde{\mu}\, g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}] + \frac{1}{12} m_1\tilde{\mu}\, g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde{d}}^{p}, \tilde{\mu}] - \\
&\frac{1}{12} m_1\tilde{\mu}\, g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde{d}}^{p}, \tilde{\mu}] - \frac{1}{144} m_1\tilde{\mu}\, g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{q}}^{i2}, m_{\tilde{d}}^{p}] + \\
&\frac{1}{36} m_1\tilde{\mu}\, g_1^2\, \overline{y_u}^{i2p} y_u^{i1p}\, \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}] - \frac{1}{72} m_1\tilde{\mu}\, g_1^2\, \overline{y_u}^{i2p} y_u^{i1p} \\
&\quad \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}] - \frac{1}{36} m_1\tilde{\mu}\, g_1^2\, \overline{y_u}^{i2p} y_u^{i1p}\, \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}] + \\
&\frac{1}{72} m_1\tilde{\mu}\, g_1^2\, \overline{y_u}^{i2p} y_u^{i1p}\, \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{u}}^{p}, m_{\tilde{q}}^{i2}] - \frac{1}{6} m_1\tilde{\mu}\, g_1^2\, \overline{y_u}^{i2p} y_u^{i1p}\, \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde{u}}^{p}, \tilde{\mu}] + \\
&\frac{1}{6} m_1\tilde{\mu}\, g_1^2\, \overline{y_u}^{i2p} y_u^{i1p}\, \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde{u}}^{p}, \tilde{\mu}] - \frac{5}{48} m_1\tilde{\mu}\, g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, \mathrm{LF}_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{d}}^{p}] - \\
&\frac{1}{48} m_1\tilde{\mu}\, g_1^2\left(\overline{y_d}^{i2p} y_d^{i1p} + \overline{y_u}^{i2p} y_u^{i1p}\right) \mathrm{LF}_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{q}}^{i2}] + \\
&\frac{7}{48} m_1\tilde{\mu}\, g_1^2\, \overline{y_u}^{i2p} y_u^{i1p}\, \mathrm{LF}_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{u}}^{p}] - \frac{3}{16} m_2\tilde{\mu}\, g_2^2\, \overline{y_d}^{i2p} y_d^{i1p}\, \mathrm{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_{\tilde{d}}^{p}] - \\
&\frac{3}{16} m_2\tilde{\mu}\, g_2^2\left(\overline{y_d}^{i2p} y_d^{i1p} - \overline{y_u}^{i2p} y_u^{i1p}\right) \mathrm{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_{\tilde{q}}^{i2}] + \\
&\frac{3}{16} m_2\tilde{\mu}\, g_2^2\, \overline{y_u}^{i2p} y_u^{i1p}\, \mathrm{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_{\tilde{u}}^{p}] - \frac{1}{3} m_3\tilde{\mu}\, g_3^2\, \overline{y_d}^{i2p} y_d^{i1p}\, \mathrm{LF}_{2,1,1,0}[m_3, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}] - \\
&\frac{1}{6} m_3\tilde{\mu}\, g_3^2\, \overline{y_d}^{i2p} y_d^{i1p}\, \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}] + \frac{1}{3} m_3\tilde{\mu}\, g_3^2\, \overline{y_d}^{i2p} y_d^{i1p}\, \mathrm{LF}_{3,1,1,-1}[m_3, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}] + \\
&\frac{1}{6} m_3\tilde{\mu}\, g_3^2\, \overline{y_d}^{i2p} y_d^{i1p}\, \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde{q}}^{i2}, m_{\tilde{d}}^{p}] + \frac{1}{3} m_3\tilde{\mu}\, g_3^2\, \overline{y_u}^{i2p} y_u^{i1p}\, \mathrm{LF}_{2,1,1,0}[m_3, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}] - \\
&\frac{1}{6} m_3\tilde{\mu}\, g_3^2\, \overline{y_u}^{i2p} y_u^{i1p}\, \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}] - \frac{1}{3} m_3\tilde{\mu}\, g_3^2\, \overline{y_u}^{i2p} y_u^{i1p}\, \mathrm{LF}_{3,1,1,-1}[m_3, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}] + \\
&\frac{1}{6} m_3\tilde{\mu}\, g_3^2\, \overline{y_u}^{i2p} y_u^{i1p}\, \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde{u}}^{p}, m_{\tilde{q}}^{i2}] - \frac{1}{16}\tilde{\mu}\, \overline{y_d}^{i2p}\, \overline{y_u}^{rs} y_u^{i1s} a_d^{rp} \\
&\quad \mathrm{LF}_{2,2,1,-1}[m_{\tilde{d}}^{p}, \tilde{\mu}, m_{\tilde{q}}^{r}] - \frac{1}{16}\tilde{\mu}\, \overline{y_d}^{pr} y_d^{i1r}\, \overline{y_u}^{i2s} a_u^{ps}\, \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{p}, \tilde{\mu}, m_{\tilde{u}}^{s}] + \\
&\frac{1}{16}\tilde{\mu}\, \overline{y_d}^{i2p}\, \overline{y_u}^{rs} y_u^{i1s} a_d^{rp}\, \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{r}, \tilde{\mu}, m_{\tilde{d}}^{p}] + \frac{1}{16}\tilde{\mu}\, \overline{y_d}^{pr} y_d^{i1r}\, \overline{y_u}^{i2s} a_u^{ps} \\
&\quad \mathrm{LF}_{2,2,1,-1}[m_{\tilde{u}}^{s}, \tilde{\mu}, m_{\tilde{q}}^{p}] + \frac{1}{24} m_1\tilde{\mu}\, g_1^2\left(\overline{y_d}^{i2p} y_d^{i1p} + \overline{y_u}^{i2p} y_u^{i1p}\right) \mathrm{LF}_{2,1,1,0}[\tilde{\mu}, m_1, m_{\tilde{q}}^{i2}] - \\
&\frac{1}{24} m_1\tilde{\mu}\, g_1^2\left(\overline{y_d}^{i2p} y_d^{i1p} + \overline{y_u}^{i2p} y_u^{i1p}\right) \mathrm{LF}_{3,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{q}}^{i2}] + \\
&\frac{3}{8} m_2\tilde{\mu}\, g_2^2\left(\overline{y_d}^{i2p} y_d^{i1p} - \overline{y_u}^{i2p} y_u^{i1p}\right) \mathrm{LF}_{2,1,1,0}[\tilde{\mu}, m_2, m_{\tilde{q}}^{i2}] + \\
&\frac{3}{8} m_2\tilde{\mu}\, g_2^2\left(-\overline{y_d}^{i2p} y_d^{i1p} + \overline{y_u}^{i2p} y_u^{i1p}\right) \mathrm{LF}_{3,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{q}}^{i2}] + \\
&\frac{1}{8}\tilde{\mu}\, \overline{y_d}^{i2p}\, \overline{y_u}^{rs} y_u^{i1s} a_d^{rp}\, \mathrm{LF}_{2,1,1,0}[\tilde{\mu}, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{r}] - \\
&\frac{1}{8}\tilde{\mu}\, \overline{y_d}^{i2p}\, \overline{y_u}^{rs} y_u^{i1s} a_d^{rp}\, \mathrm{LF}_{3,1,1,-1}[\tilde{\mu}, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{r}] - \frac{1}{8}\tilde{\mu}\, \overline{y_d}^{pr} y_d^{i1r}\, \overline{y_u}^{i2s} a_u^{ps} \\
&\quad \mathrm{LF}_{2,1,1,0}[\tilde{\mu}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{s}] + \frac{1}{8}\tilde{\mu}\, \overline{y_d}^{pr} y_d^{i1r}\, \overline{y_u}^{i2s} a_u^{ps}\, \mathrm{LF}_{3,1,1,-1}[\tilde{\mu}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{s}] - \\
&\frac{1}{48}\tilde{\mu}\, g_1^2\, \overline{y_d}^{i2p} a_d^{i1p}\, \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}, \tilde{\mu}] + \frac{1}{96}\tilde{\mu}\, g_1^2\, \overline{y_d}^{i2p} a_d^{i1p} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}, \tilde{\mu}] - \frac{1}{48} m_1\tilde{\mu}\, g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}, \tilde{\mu}] - \\
&\frac{1}{96} m_1\tilde{\mu}\, g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}, \tilde{\mu}] - \frac{1}{48}\tilde{\mu}\, g_1^2\, \overline{y_u}^{i2p} a_u^{i1p} \\
&\quad \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}, \tilde{\mu}] + \frac{1}{96}\tilde{\mu}\, g_1^2\, \overline{y_u}^{i2p} a_u^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}, \tilde{\mu}] - \\
&\frac{1}{48} m_1\tilde{\mu}\, g_1^2\, \overline{y_u}^{i2p} y_u^{i1p}\, \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}, \tilde{\mu}] - \frac{1}{96} m_1\tilde{\mu}\, g_1^2\, \overline{y_u}^{i2p} y_u^{i1p} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}, \tilde{\mu}] - \frac{3}{16}\tilde{\mu}\, g_2^2\, \overline{y_d}^{i2p} a_d^{i1p}\, \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}, \tilde{\mu}] + \\
&\frac{3}{32}\tilde{\mu}\, g_2^2\, \overline{y_d}^{i2p} a_d^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}, \tilde{\mu}] - \frac{3}{16} m_2\tilde{\mu}\, g_2^2\, \overline{y_d}^{i2p} y_d^{i1p} \\
&\quad \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}, \tilde{\mu}] - \frac{3}{32} m_2\tilde{\mu}\, g_2^2\, \overline{y_d}^{i2p} y_d^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}, \tilde{\mu}] + \\
&\frac{3}{16}\tilde{\mu}\, g_2^2\, \overline{y_u}^{i2p} a_u^{i1p}\, \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}, \tilde{\mu}] - \frac{3}{32}\tilde{\mu}\, g_2^2\, \overline{y_u}^{i2p} a_u^{i1p} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}, \tilde{\mu}] + \frac{3}{16} m_2\tilde{\mu}\, g_2^2\, \overline{y_u}^{i2p} y_u^{i1p}\, \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}, \tilde{\mu}] + \\
&\frac{3}{32} m_2\tilde{\mu}\, g_2^2\, \overline{y_u}^{i2p} y_u^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}, \tilde{\mu}] - \frac{1}{144}\tilde{\mu}\, g_1^2\, \overline{y_d}^{i2p} a_d^{i1p} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{d}}^{p}, m_1, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}] - \frac{1}{96}\tilde{\mu}\, g_1^2\, \overline{y_d}^{i2p} a_d^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{d}}^{p}, m_1, m_{\tilde{q}}^{i1}, \tilde{\mu}] + \\
&\frac{1}{96} m_1\tilde{\mu}\, g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{d}}^{p}, m_1, m_{\tilde{q}}^{i2}, \tilde{\mu}] - \frac{3}{16}\tilde{\mu}\, g_2^2\, \overline{y_d}^{i2p} a_d^{i1p} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{d}}^{p}, m_2, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}] - \frac{3}{32}\tilde{\mu}\, g_2^2\, \overline{y_d}^{i2p} a_d^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{d}}^{p}, m_2, m_{\tilde{q}}^{i1}, \tilde{\mu}] + \\
&\frac{3}{32} m_2\tilde{\mu}\, g_2^2\, \overline{y_d}^{i2p} y_d^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{d}}^{p}, m_2, m_{\tilde{q}}^{i2}, \tilde{\mu}] - \frac{1}{3}\tilde{\mu}\, g_3^2\, \overline{y_d}^{i2p} a_d^{i1p} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{d}}^{p}, m_3, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}] + \frac{1}{4}\tilde{\mu}\, \overline{y_d}^{rs}\, \overline{y_d}^{i2p} y_d^{i1s} a_d^{rp}\, \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{q}}^{r}, m_{\tilde{d}}^{p}, m_{\tilde{d}}^{s}, \tilde{\mu}] - \\
&\frac{1}{4}\tilde{\mu}\, \overline{y_u}^{rs}\, \overline{y_u}^{i2p} y_u^{i1s} a_u^{rp}\, \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{q}}^{r}, m_{\tilde{u}}^{p}, m_{\tilde{u}}^{s}, \tilde{\mu}] + \frac{1}{96}\tilde{\mu}\, g_1^2\, \overline{y_d}^{i2p} a_d^{i1p} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{q}}^{i1}, m_1, m_{\tilde{d}}^{p}, \tilde{\mu}] + \frac{1}{96}\tilde{\mu}\, g_1^2\, \overline{y_u}^{i2p} a_u^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{q}}^{i1}, m_1, m_{\tilde{u}}^{p}, \tilde{\mu}] + \\
&\frac{3}{32}\tilde{\mu}\, g_2^2\, \overline{y_d}^{i2p} a_d^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{q}}^{i1}, m_2, m_{\tilde{d}}^{p}, \tilde{\mu}] - \frac{3}{32}\tilde{\mu}\, g_2^2\, \overline{y_u}^{i2p} a_u^{i1p} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{q}}^{i1}, m_2, m_{\tilde{u}}^{p}, \tilde{\mu}] - \frac{1}{96} m_1\tilde{\mu}\, g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{q}}^{i2}, m_1, m_{\tilde{d}}^{p}, \tilde{\mu}] - \\
&\frac{1}{96} m_1\tilde{\mu}\, g_1^2\, \overline{y_u}^{i2p} y_u^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{q}}^{i2}, m_1, m_{\tilde{u}}^{p}, \tilde{\mu}] - \frac{3}{32} m_2\tilde{\mu}\, g_2^2\, \overline{y_d}^{i2p} y_d^{i1p} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{q}}^{i2}, m_2, m_{\tilde{d}}^{p}, \tilde{\mu}] + \frac{3}{32} m_2\tilde{\mu}\, g_2^2\, \overline{y_u}^{i2p} y_u^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{q}}^{i2}, m_2, m_{\tilde{u}}^{p}, \tilde{\mu}] + \\
&\frac{1}{144}\tilde{\mu}\, g_1^2\, \overline{y_u}^{i2p} a_u^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{u}}^{p}, m_1, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}] - \frac{1}{96}\tilde{\mu}\, g_1^2\, \overline{y_u}^{i2p} a_u^{i1p} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{u}}^{p}, m_1, m_{\tilde{q}}^{i1}, \tilde{\mu}] + \frac{1}{96} m_1\tilde{\mu}\, g_1^2\, \overline{y_u}^{i2p} y_u^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{u}}^{p}, m_1, m_{\tilde{q}}^{i2}, \tilde{\mu}] + \\
&\frac{3}{16}\tilde{\mu}\, g_2^2\, \overline{y_u}^{i2p} a_u^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{u}}^{p}, m_2, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}] + \frac{3}{32}\tilde{\mu}\, g_2^2\, \overline{y_u}^{i2p} a_u^{i1p} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{u}}^{p}, m_2, m_{\tilde{q}}^{i1}, \tilde{\mu}] - \frac{3}{32} m_2\tilde{\mu}\, g_2^2\, \overline{y_u}^{i2p} y_u^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{u}}^{p}, m_2, m_{\tilde{q}}^{i2}, \tilde{\mu}] + \\
&\frac{1}{3}\tilde{\mu}\, g_3^2\, \overline{y_u}^{i2p} a_u^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{u}}^{p}, m_3, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}] - \frac{1}{96}\tilde{\mu}\, g_1^2\, \overline{y_d}^{i2p} a_d^{i1p} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}] + \frac{1}{96} m_1\tilde{\mu}\, g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}] - \\
&\frac{1}{96}\tilde{\mu}\, g_1^2\, \overline{y_u}^{i2p} a_u^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}] + \frac{1}{96} m_1\tilde{\mu}\, g_1^2\, \overline{y_u}^{i2p} y_u^{i1p} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}] - \frac{3}{32}\tilde{\mu}\, g_2^2\, \overline{y_d}^{i2p} a_d^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}] + \\
&\frac{3}{32} m_2\tilde{\mu}\, g_2^2\, \overline{y_d}^{i2p} y_d^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}] + \frac{3}{32}\tilde{\mu}\, g_2^2\, \overline{y_u}^{i2p} a_u^{i1p} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}] - \frac{3}{32} m_2\tilde{\mu}\, g_2^2\, \overline{y_u}^{i2p} y_u^{i1p}\, \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}]\Bigg)
\end{aligned}
$$